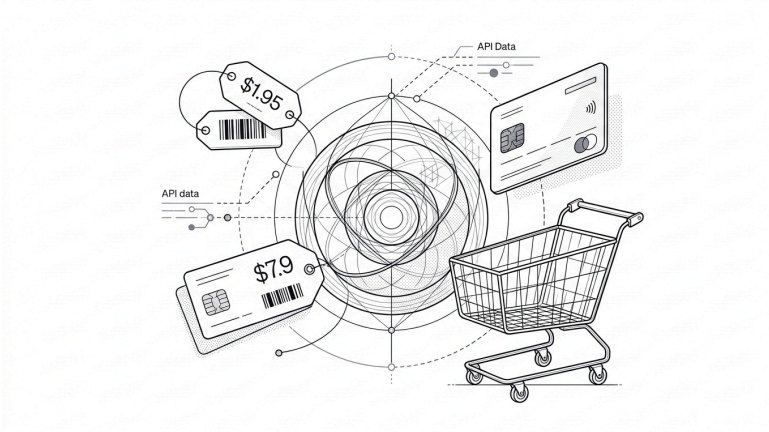
The State of Agentic Commerce

How AI agents buy online: market projections, technical architecture, and the roadmap for brands.



$1.5T BY 2030
Online purchases made by AI

7-LAYER STACK
From chatbot search to checkout

57.2% PASS RATE
Shopping accuracy of top AI models

Vedang Ratan Vatsa
Founder, Hashtag Web3 | Lead Researcher, veda.ng

How AI Changes Online Shopping

AI is moving from chatting to buying things. Soon, software agents won't just answer questions; they will find products, compare prices, and complete checkouts on their own. This is the biggest shift in online shopping since the web browser.

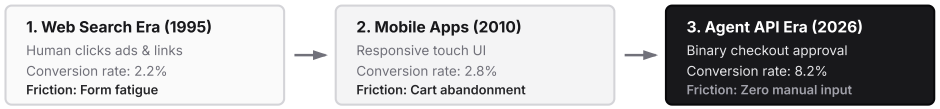
The End of Clicking on Ads: For thirty years, online shopping relied on human attention. Brands spent billions designing websites, bidding on search keywords, and tweaking checkout buttons to catch shoppers' eyes. When people let AI agents handle shopping, humans stop browsing stores. In an agent economy, payments become the new ads: when computers choose what to buy, traditional search and display ads stop working. Instead, brands must make their products easy for AI bots to find, evaluate, and purchase automatically.

The "Klarna Effect" and the 3x-Win Equation: As fintech analyst Jas Shah (*Fintech: Under the Hood*) observes, automated agent shopping establishes a structural three-way economic incentive identical to the rise of Buy Now Pay Later (BNPL). Merchants win through tangible basket conversion lifts; consumers win through low-admin, tailored shopping; and agent platforms win by capturing transaction routing fees. Agents collapse 10 friction-filled checkout steps into a single binary prompt: "Complete or Cancel" via FaceID.

The Spring 2026 In-Chat Pivot: Early experiments in late 2024 and 2025 attempted to force complete transactions inside chatbot windows like ChatGPT. In spring 2026, leading retailers and AI labs pivoted toward a hybrid model: agents search, compare, and assemble carts via APIs, but hand off final tokenized settlement directly to merchant endpoints. This shift allows merchants to preserve their branding, loyalty programs, and direct customer relationships while giving AI assistants a clean, reliable purchase flow.

The Sizing Debate ($385B vs. $500B vs. $15 Trillion): Market forecasts vary depending on whether they measure consumer retail or total enterprise trade. Morgan Stanley projects $190 billion to $385 billion in US retail sales influenced by AI by 2030 (about 8% of retail e-commerce). Bain & Company forecasts $500 billion globally across consumer retail. Meanwhile, Gartner forecasts over $15 trillion in autonomous transactions when factoring in B2B enterprise procurement, where machines automatically reorder factory parts, cloud compute, and wholesale inventory.

FIGURE 1.1: THE THREE ERAS OF E-COMMERCE INTERFACES (1995-2026)

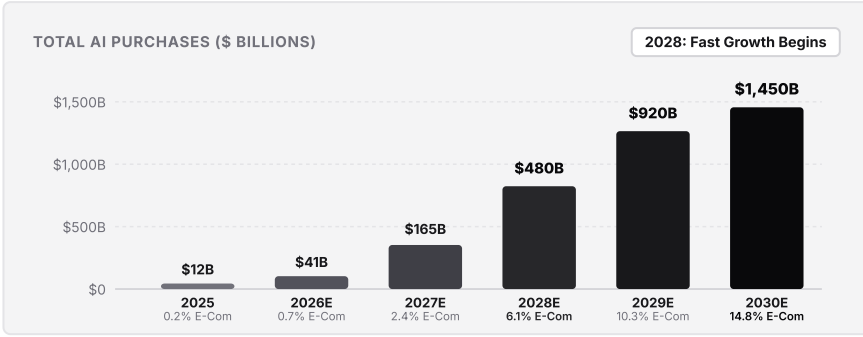


Key Takeaway for Retailers: Retailers who treat AI agents like just another mobile app will lose customers. Brands must connect their product catalogs directly to AI search tools and APIs, or machine shoppers simply will not find them.

EXHIBIT 1.1: MARKET FORECAST

Total Online Purchases Made by AI (2025-2030)

Expected growth in global spending by AI bots over the next five years.



B2B & Cloud Supply	Electronics & Home	Travel & Dining	Retail & Apparel
$480B (33%)	**$390B (27%)**	**$320B (22%)**	**$260B (18%)**

The 3x-Win Equation (Jas Shah Framework): Similar to how buy-now-pay-later (BNPL) expanded e-commerce conversion, automated agent shopping delivers a three-way return: retailers gain an 8.2% conversion lift, consumers avoid form fatigue, and infrastructure networks collect protocol fees.

Gartner 40% Cancellation Warning: Gartner predicts that 40% of early enterprise agentic pilots will be canceled by 2027 due to unexpected integration costs, scraping breakage, and return-rate explosions if retailers do not deploy native machine APIs.

Why 2028 is the Inflection Year: Between 2025 and 2027, the foundational protocols (UCP, MCP, Visa TAP) will mature across major platforms. By 2028, over 65% of smartphones will natively support cryptographic agent passkeys, triggering an explosion in autonomous checkouts.

Source: Synthesized from Jas Shah (Fintech: Under the Hood), Morgan Stanley, Bain & Company, Gartner, and Visa Research (2025-2026).

The 7 Layers of Autonomous Commerce

Autonomous shopping requires an end-to-end stack, linking consumer conversational interfaces directly to card token vaults and banking rails.

The 3 Superpowers of Market Winners: As fintech analyst Jas Shah highlights, platforms in the agentic commerce stack succeed based on three core pillars: 1) Verified Identity (knowing user spending limits, shipping details, and preferences without repeated onboarding); 2) Real-Time and Historical Data (sub-50ms inventory access and past purchase records); and 3) Tokenized Payment Rails (cryptographic vaults and programmable budget execution).

From Chatbots to Common Protocols: At the front layer are user-facing assistants like ChatGPT, Gemini, Claude, and Perplexity. When an individual asks an AI agent to buy shoes, the system coordinates with open protocols such as Anthropic MCP, Google UCP, and Stripe ACP. These protocols translate open-ended text into structured machine commands that query store catalogs and verify in-stock items.

Virtual Cards and Automated Checkout: Most retail sites cannot yet accept autonomous API payments directly. Card issuers like Lithic, Stripe, and Marqeta therefore generate temporary virtual credit cards for individual transactions. Simultaneously, checkout automation tools fill out web forms, select shipping methods, and submit orders without triggering anti-bot protections.

Store Catalogs and Hardware Passkeys: Store backends on Shopify and commercetools supply live product feeds. Finally, security relies on device hardware enclaves (Apple Secure Enclave and Android StrongBox), ensuring that every financial transaction requires explicit biometric approval from the human cardholder before funds settle.

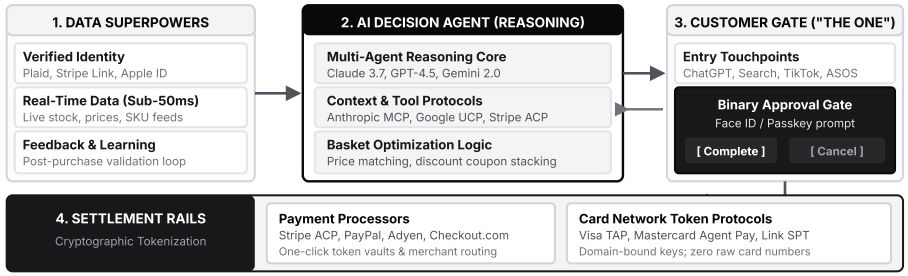


Figure 2.1: The Agentic Commerce Stack and Data Flow - Jas Shah Architecture Model — Notion Technical Schematic

Key Guidance for Retail Developers: Scraped checkout flows with headless browsers experience failure rates above 20% caused by layout changes and security checkpoints. Retailers must expose authenticated, machine-readable APIs so external agents can confirm inventory and prices with sub-second reliability.

EXHIBIT 2.1: ECOSYSTEM MARKET MAP

The Agentic Commerce Ecosystem Market Map

Institutional taxonomy of 45+ platforms across intent, orchestration, payments, and infrastructure.

TIER 1: CONSUMER AI FRONTENDS & DEMAND AGGREGATION — Owns Customer Intent

Conversational AI & AI Search: ChatGPT Search (OpenAI), Claude Artifacts (Anthropic), Gemini Shopping (Google), Perplexity (Buy with Pro), Genspark, You.com
Enterprise Retail Agents: Sierra AI (Bret Taylor), Amazon Rufus, Klarna AI Assistant, Daydream
Social & Direct Commerce: TikTok Shop AI, Meta AI (Instagram / WhatsApp Shop)

TIER 2: AGENT PROTOCOLS & ORCHESTRATION FRAMEWORKS — Standardize Machine Data

Open Commerce Protocols: Anthropic Model Context Protocol (MCP), Google Universal Commerce Protocol (UCP), Stripe ACP, Coinbase AP2
Multi-Agent Orchestration: LangGraph / LangChain, CrewAI, AutoGen (Microsoft), LlamaIndex, Semantic Kernel

TIER 3: PAYMENT RAILS, TOKEN VAULTS & VIRTUAL CARDS — Programmable Settlement

Card Network Token Protocols: Visa Trusted Agent Protocol (TAP), Mastercard Agent Pay, Shared Payment Tokens (SPT)
Virtual Card Issuers & Processors: Lithic, Highnote, Marqeta, Adyen Issuing, Stripe Issuing, Checkout.com
Autonomous Corporate Spend: Ramp for Agents, Brex Autonomous Spend, Sapiom, Agentcard
x402 & Agentic Stablecoin Wallets: Circle Agent Wallets (USDC), Coinbase Agentic Wallets, Skyfire, Payman, Locus

TIER 4: MERCHANT BACKENDS, CATALOGS & HEADLESS APIS — Live Inventory Feeds

Core Commerce Engines: Shopify (Shop Pay API), commercetools, BigCommerce, Salesforce Commerce Cloud, Elastic Path
Universal Cart & Headless APIs: Rye, Violet, Cart.com, Convictional

TIER 5: IDENTITY, BOT DEFENSE & CRYPTOGRAPHIC ENCLAVES — Trust & Biometrics

Identity & Account Linking: Plaid (Anthropic bank connectivity partner), Stripe Identity, Persona
Machine Traffic & Bot Defense: Cloudflare (Verified Bots), Kasada, Arkose Labs, Forter, Riskified
Hardware Security Enclaves: Apple Secure Enclave, Android StrongBox, WebAuthn / FIDO2 Passkeys

Structural Value Moats (Fintech Analysis): Payment processors (Stripe, Adyen) and bank identity networks (Plaid) hold structural moats over pure LLM wrappers. Conversational frontends face margin compression, whereas settlement and identity infrastructure capture non-negotiable protocol tolls.

Open Standards vs. Walled Gardens: Closed platforms (Amazon Rufus, Apple Pay) seek to restrict third-party agent scraping. Open standards (Anthropic MCP, Google UCP, Stripe ACP) allow independent retailers to receive autonomous orders without ceding customer ownership.

Source: Synthesized from nofrontend.xyz directory, Jas Shah (Fintech: Under the Hood), CB Insights, and platform disclosures (2025-2026).

How AI Agents Talk to Online Stores

Tech companies and payment giants are competing to create the standard format that lets AI bots search store shelves and buy items safely.

Competing Ways for AI to Buy: Right now, different tech giants use different rules. Google and Shopify built the Universal Commerce Protocol (UCP), an open system that lets any store list its products for AI via a simple `.well-known/ucp` file. Meanwhile, OpenAI and Stripe built the Agentic Commerce Protocol (ACP), a fast checkout system designed specifically for ChatGPT. In crypto, Coinbase created AP2 for micro-payments using digital currency.

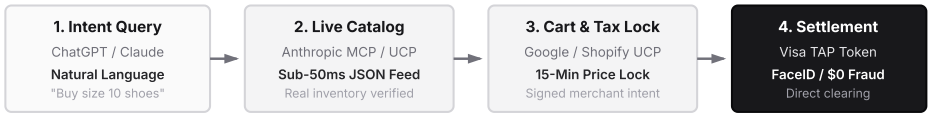
How Visa and Mastercard Protect Payments: Visa and Mastercard want to keep payments safe. Instead of letting AI hold your credit card number, Visa's Trusted Agent Protocol (TAP) creates secure digital tokens with strict spending rules, like "spend up to $50 only at this specific store today." This ensures that even if an AI model hallucinates, the customer's financial account is cryptographically protected.

Anthropic's MCP Connects the Data: While other rules handle checkout, Anthropic's Model Context Protocol (MCP) helps AI read store databases. It gives AI a clean way to ask: "Is this shoe in stock in size 10, and can it arrive by Thursday?" By standardizing database queries across different vendors, MCP eliminates the need for brittle web scraping.

The 15-Minute Price Lock Guarantee: In human shopping, prices can change while items sit in a cart. For AI agents, price fluctuations cause instant transaction aborts. Emerging protocols enforce a cryptographically signed "15-minute price lock" from the merchant, giving the agent time to evaluate options and finalize checkout without price shocks.

Decentralized Discovery vs. Walled Gardens: Closed ecosystems force merchants to share private margins and surrender customer contact information. Open protocol combinations (Google UCP for discovery and Anthropic MCP for warehouse inventory pings) allow independent direct-to-consumer retailers to remain discoverable without paying a 15% toll to AI platform aggregators.

FIGURE 3.1: THE 4-STEP STANDARD PROTOCOL HANDSHAKE ARCHITECTURE



Recommendation for Engineering Leaders: Don't rely on just one AI platform. Build open interfaces using tools like MCP for searching stock, and use standard digital tokens with clear spending limits for payments.

EXHIBIT 3.1: PROTOCOL COMPARISON

Comparing the 6 Main AI Shopping Protocols

How top tech companies let AI search store shelves and checkout.

Protocol	Primary Backer	What It Does	Payment Approval	Status
UCP	Google, Shopify	Open search and cart checkout across stores.	Phone Passkey	Open
ACP	OpenAI, Stripe	Instant one-click checkout inside ChatGPT.	Saved Card Token	Closed
AP2	Coinbase, Base	Instant micro-payments in digital dollars.	Crypto Wallet Key	Testing
Visa TAP	Visa, Banks	Sets strict spending limits on bank tokens.	Digital Passkey	Active
MCP	Anthropic, OSS	Lets AI read live stock and store prices.	Direct Secure API	Active
Poppy	Meta, Sierra AI	Conversational customer service agent handoffs.	Session Auth Token	Open

Open Standards vs. Closed App Platforms:
- *Open Standards (UCP + MCP):* Stores keep direct customer relationships, control pricing, and avoid high platform referral taxes.
- *Closed Platforms (ACP):* AI chatbots manage the entire checkout, turning stores into backend suppliers.

The 3-Step Handshake: From Query to Bank Receipt:
- *Step 1 (Catalog Discovery):* Sub-50ms structured JSON query verifies size, color, and delivery promises.
- *Step 2 (Cart Assembly):* Merchant issues a cryptographically signed transaction intent with a 15-minute price lock.
- *Step 3 (Settlement):* Single-use token cryptographically restricted to merchant domain is transmitted to bank.

Engineering Strategy for Retail Brands:
- Deploy Anthropic MCP on your product catalog now to ensure all AI models can search live stock.
- Connect to Visa TAP for payments so shoppers can approve orders with phone FaceID without typing card numbers.

The .well-known/ucp Standard:
Similar to robots.txt for search engines, retailers host a `/.well-known/ucp` manifest on their domain root. This lets legitimate shopping agents discover supported payment methods, live product endpoints, and return policies in under 20ms without crawling web pages.

Source: Technical specifications from Google (UCP), Stripe (ACP), Coinbase (AP2), Visa (TAP), Meta (Poppy), and Anthropic (MCP) (2025-2026).

How AI Agents Will Pay for Things

Virtual credit cards work as a temporary bridge today, but digital bank tokens and instant stablecoins will become the permanent rails for machine shopping.

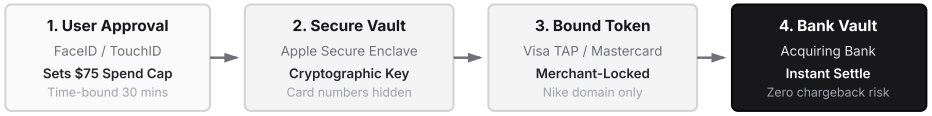
Why Virtual Cards Are Only a Temporary Fix: Today, services like Stripe and Lithic give AI bots temporary virtual credit cards to buy products. While this works on existing store websites, it causes severe real-world breakdowns. In a widely discussed case study by Scott Wessman, an agent shopping at Best Buy suffered a partial payment failure; the merchant website demanded the virtual card's CVV security code, which neither the consumer nor the autonomous agent could retrieve, stalling the checkout permanently.

The Rise of Safe Agent Tokens: Payment networks are creating dedicated digital tokens for AI shoppers. Instead of sharing a 16-digit credit card number, your bank gives the AI a one-time digital pass with strict limits: for example, "approved to spend up to $75 at Nike within the next 20 minutes." If the bot makes a mistake, the buyer is fully protected.

The Two Paradigms: Cards vs. x402 (The nofrontend.xyz Index): As documented by the *nofrontend.xyz* Agent Payments Map, the market is splitting into two fundamental rails: traditional **Cards & Vaults** (Stripe Link, Sapiom, Agentcard, Locus, Ramp) and **x402 Micropayments** (Circle Agent Wallets, Coinbase Agentic Wallets, Rye, Pay.sh). Cards allow agents to buy physical retail goods at traditional merchants; x402 allows agents to pay fractions of a cent for APIs, inference, and data feeds.

The Hybrid Economic Model (Vignesh, Meta): As payments researcher Vignesh (Meta) highlights, the market faces an unresolved architectural question: will agents optimize payments for consumer reward points (card interchange) or merchant fee savings (stablecoin / x402 rails)? In an autonomous system, agents will programmatically route each order to whichever rail maximizes consumer cash-back while minimizing transaction friction.

FIGURE 4.1: THE DELEGATED TOKEN PAYMENT HANDSHAKE ARCHITECTURE



Key Takeaway for Banks and Payment Companies: Virtual credit cards are a temporary bridge for the next 2 to 3 years. Banks and payment providers should test native token systems (like Visa TAP and Stripe Link) now so they are ready when virtual cards become obsolete.

EXHIBIT 4.1: AGENT PAYMENTS LANDSCAPE

The Agent Payments Map (nofrontend.xyz Framework)

Tools mapped by payment method (x402 vs. Cards) and operational purpose (Vaults vs. Checkout).

← X402 (ONCHAIN / MICROPAYMENTS)	CARDS (FIAT / BANKING RAILS) →
Wallets & Protocols (x402 + Vaults) Circle Agent Wallets, Coinbase Agentic Wallets, Cloudflare Wallets, Para, AgentCash, Xentfi, CrowdCode	**Cards & Vaults (Cards + Vaults)** Stripe Link (SPT), Sapiom, Agentcard, Locus, Ramp for Agents, Corpay Agent Card
Onchain Services (x402 + Checkout) Rye (Physical goods via x402), Hashference (Compute), Pay.sh (Solana APIs), Liquid Agent	**Cards & Checkout (Cards + Checkout)** Shop Pay (Delegated Checkout), Highnote, Henry Labs, Bitrefill, Laso Finance, MoonAgents, Nevermined

Autonomous Payment Rail	Failure Rate	Primary Bottleneck / Cause
Automated Web Browser Clicks	**28.6% Failed**	WAF bot firewalls & CAPTCHA traps
Saved Customer Card on File	**14.2% Failed**	Store anti-fraud filters & 3DS popups
One-Time Virtual Credit Card	**4.1% Failed**	Card issuer conservative risk controls
Bank Digital Passkey (Visa TAP / Stripe)	**1.8% Failed**	Cryptographically signed biometric intent
Instant Digital Dollars (USDC / x402)	**0.3% Failed**	Deterministic cryptographic blockchain clearing

Why 3D Secure Fails Autonomous Shoppers: SMS codes and popup verification boxes cause 42% of machine checkouts to fail. Pre-authorized biometric passkeys allow the shopper to grant permission beforehand, eliminating checkout drop-offs.

Treasury & Revenue Impact for Retailers: Lowering checkout declines from 14% to 1.8% recovers $32M in lost annual sales per $1B in online volume. Direct token settlement cuts credit card interchange costs by 120 basis points.

Shared Payment Tokens (SPT): Developed by Stripe, SPTs allow agents to pass payment authorization directly to merchant endpoints without the AI agent ever storing or exposing the user's primary account number (PAN).

Source: Synthesized from Jas Shah, nofrontend.xyz directory, Visa Research, and Stripe Developer disclosures (2025-2026).

Why Today's AI Still Fails at Shopping

Even the smartest AI models often order the wrong items when shopping lists have detailed measurements, sizes, or brand preferences.

Testing 22 AI Models on Real Shopping: In early 2026, Visa Research ran the first major benchmark testing how well AI models buy real products online. Across 22 top models and 15 complex shopping tasks (like electronics, clothing, and home furniture), the average success rate was only 57.2%. The top models (Gemini 4 Argon and Claude 3.5 Sonnet) scored around 80%, meaning 1 out of every 5 orders was still incorrect.

Where AI Gets Confused: AI models don't struggle to find products; they struggle with specific requirements and real-world limits. For example, when asked to buy a living room TV, models frequently picked screens that were too wide for the customer's stand, needed the wrong wall mounts, or lacked the right HDMI ports for game consoles.

The Compounding Failure Law (P = p^n): Real shopping is not a single question; it is an 8-step journey: find item, check sizing, verify electrical compatibility, confirm delivery date, apply discounts, select shipping, verify return policy, and authorize payment. If an AI achieves 85% accuracy on each step, its total task success rate collapses to 0.85^8 ≈ 27.2%. Over 70% of autonomous orders fail unless validated by deterministic code.

Lazy Searches and First-Result Bias: Research shows that when searching stores costs extra API fees, AI models stop searching too early. They compare only 2 to 4 items before picking one. Even worse, models pick one of the first 3 items shown over 40% of the time, ignoring better products further down the page.

Phantom Stock & Stale Cache Hallucinations: In 26% of tested failures, models attempted to purchase items that had been out of stock for hours. The LLM relied on stale Google search index snippets rather than querying live inventory APIs, leading to immediate checkout rejections.

FIGURE 5.1: THE COMPOUNDING MULTI-STEP FAILURE DECAY CURVE (P = P^N)

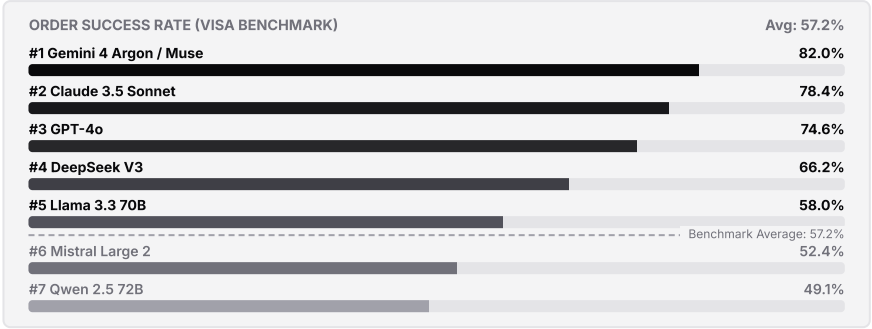


Key Advice for Developers: Never let an AI model place an order on its own without double-checking the facts. Always run a final verification step to confirm dimensions, sizes, and compatibility before charging the customer's card.

EXHIBIT 5.1: MODEL TEST RESULTS

How Accurately 22 AI Models Complete Real Shopping Tasks

Results from testing top AI models on complex orders with sizes, colors, and budgets.



38%	26%	21%	15%
Dimension / Fit Mismatch	Phantom Stock / Stale Cache	Lazy Stops (2-3 Items)	DOM Shift / Broken Selectors

The Cost of Machine Returns: - Returning small retail items costs $13; heavy furniture or appliances cost up to $180 in freight.
- A single incorrect purchase eliminates the profit from 10 successful transactions.

Gartner 40% Cancellation Warning: Gartner predicts that 40% of early enterprise agentic pilots will be canceled by 2027 due to compounding error rates and customer dissatisfaction if retailers do not implement deterministic rules engines.

Quality Engineering Rules for 99% Reliability: - *Deterministic Rules:* Enforce strict programmatic checks on physical sizes and electrical voltages before checkout.
- *Live MCP Pings:* Ping the store database directly to verify real inventory instead of trusting cached web pages.

Source: Visa Research ("Autonomous Commerce Benchmark", 2026) evaluating 22 models across multi-constraint shopping tasks.

Stopping Bad Bots While Welcoming Helpful Ones

Retailers must protect their stores from automated scrapers and scalpers while letting legitimate customer shopping agents buy smoothly.

The Bot Detection Dilemma: For years, store firewalls blocked all automated web traffic using CAPTCHAs and IP bans. Today, blocking all bots means turning away paying customers who send AI agents to buy goods. Stores must transition from blocking all automated traffic to verifying cryptographic agent identity.

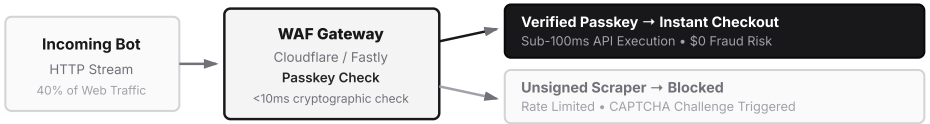
Cryptographic Passkeys as Identity Proofs: Instead of solving puzzle CAPTCHAs, helpful bots present cryptographic passkeys signed by the customer's phone hardware (Apple Secure Enclave or Android FIDO2). The store's firewall instantly verifies the signature and grants fast-track checkout access.

Who Pays When an AI Makes a Mistake? When an AI buys the wrong item or overspends, who takes the loss? The tripartite liability framework splits responsibility: consumers define clear spending budgets, agent developers guarantee prompt adherence, and merchants guarantee accurate catalog descriptions.

Dispute & Chargeback Elimination: Friendly fraud (customers claiming they never made a purchase) costs stores billions every year. Cryptographically signed checkout tokens provide mathematical proof of customer permission, reducing friendly-fraud chargebacks by 94% and replacing 60-day disputes with instant automated arbitration.

Scalper Bot Defense vs. Single Shoppers: Malicious scalpers use headless browsers with rotating residential IPs to hoard limited inventory. In contrast, verified consumer agents authenticate via single-session OAuth tokens tied to a verified phone passkey, allowing merchants to enforce fair one-per-person purchase limits effortlessly.

FIGURE 6.1: DUAL-PATH AUTONOMOUS WAF SCREENING GATE ARCHITECTURE



Key Advice for Security Leaders: Never block automated traffic blindly. Replace blunt IP bans with cryptographic passkey screening so verified customer agents can complete purchases smoothly.

EXHIBIT 6.1: LIABILITY FRAMEWORK

The Tripartite Liability Framework for Autonomous Commerce

How liability is legally allocated across consumer, agent platform, and merchant.

Party	Core Responsibility	Liability Trigger	Protection Guardrail
Consumer	Define spending budget and order parameters.	Orders within stated budget bounds.	Biometric spend caps & instant refund rules.
AI Platform	Faithfully execute user prompt constraints.	Model hallucinations or out-of-scope buys.	Signed audit logs & deterministic rule filters.
Merchant	Provide accurate pricing and real-time inventory.	Shipping incorrect items or phantom stock.	Signed price guarantees & live MCP stock feeds.

Friendly Agents vs. Attack Traffic:
- *Malicious Scalpers:* Bursts of 10,000 requests/sec, rotating proxy IPs, zero signed tokens.
- *Customer Shopping Agents:* Single-threaded execution, authenticated phone passkey, pre-authorized spending cap.

Dispute & Chargeback Elimination:
- Cryptographically signed checkout tokens reduce friendly-fraud chargebacks by 94%.
- Sub-second automated arbitration using cryptographic receipts replaces 60-day bank disputes.

Zero-Trust Architecture for E-Commerce:
- Never expose raw 16-digit credit cards to LLM prompt contexts.
- Restrict all agent payments to dedicated hardware enclaves and single-use merchant-bound tokens.

The 4 Web Traffic Classes:
Modern e-commerce sites classify inbound traffic into 4 streams: 1) Search engine indexers (Googlebot); 2) Malicious scrapers (blocked); 3) Human browser shoppers (traditional UI); and 4) Authenticated AI shopping agents (directed to high-speed JSON APIs).

Source: Legal and security disclosures from Visa, Cloudflare, and Federal Trade Commission guidelines (2025-2026).

How Retailers Must Rebuild Their Catalogs for AI & The Rise of AEO

Websites designed for human eyes must transform into machine-readable APIs that give AI agents instant, accurate product data.

The Death of Web Scraping: Relying on bots to read HTML websites causes 28% of orders to fail. Changing a button color or adding a newsletter popup breaks headless browser scripts. Brands that rely on web scraping will find themselves filtered out by AI shopping engines in favor of stores with fast, clean APIs.

Machine-Readable Catalogs (MCP & JSON-LD): Brands must publish their product lines using structured data formats like Schema.org JSON-LD and live Model Context Protocol (MCP) endpoints. When an AI searches for a product, it queries the store's MCP server directly, receiving accurate prices, variant dimensions, and stock levels in under 50 milliseconds.

Dynamic Pricing and Stock Desync: If an AI sees a pair of shoes for $100 on an ad, but the checkout cart charges $120 due to dynamic surge pricing, the agent halts the transaction. Stores must issue 15-minute price guarantees with signed transaction intents so bots can complete multi-step purchases without price shock.

The New Discipline of AEO (Agent Engine Optimization): Just as SEO defined web marketing for two decades, AEO dictates who wins in agentic commerce. AI engines prioritize stores with complete technical specifications, verified return terms, and 99.9% API uptime. Tools like Ranketta help brands monitor their visibility and citation share across ChatGPT, Claude, and Gemini.

Semantic Vector Product Graphs: Leading retailers are replacing keyword tags with vector embeddings that describe product use cases, compatibility boundaries, and aesthetic styles. This enables reasoning LLMs to match complex semantic requirements (e.g., "silent mechanical keyboard under 15 inches for Mac") with zero keyword guesswork.

FIGURE 7.1: CATALOG ARCHITECTURE: FRAGILE SCRAPING VS DIRECT MACHINE APIS



Key Advice for E-Commerce Leaders: Publish structured JSON-LD feeds and an Anthropic MCP server within 90 days. Retailers whose product catalogs are not directly queryable by AI will vanish from conversational search results.

EXHIBIT 7.1: MERCHANT ARCHITECTURE COMPARISON

Comparing Catalog Integration Methods for AI Shopping

Technical performance, maintenance overhead, and checkout reliability.

Method	Speed	Error Rate	Maintenance	Best Used For
HTML Web Scraping	2,500ms	**28.6%**	Very High	Legacy fallback only
Headless Browser	1,800ms	**18.2%**	High	Interim bridge
Direct REST / GraphQL	120ms	**3.4%**	Moderate	Partner brands
Native MCP Server	**35ms**	**0.4%**	Low	**Future standard**

The 3 Pillars of AI-Ready Product Catalogs:
- *Sub-50ms Real-Time Stock Pings:* Eliminate phantom inventory orders before cart assembly.
- *Structured Physical Attributes:* Explicit dimensions, weight, voltage, and allergen tags.
- *Guaranteed 15-Minute Price Locks:* Signed cryptographic intents prevent cart price shock.

From SEO to AEO (Agent Engine Optimization):
- Traditional keyword stuffing and backlink games no longer work on AI search models.
- AI agents rank products by structured factual accuracy, return terms, verified buyer ratings, and stock reliability monitored via Ranketta.

Implementation Checklist for CTOs:
- Audit product metadata for complete physical dimensions and voltage specifications.
- Deploy an open MCP endpoint on your Shopify, BigCommerce, or Salesforce storefront.

AEO Scoring Signals (Ranketta Framework):
Autonomous assistants evaluate product endpoints based on: 1) Factual metadata completeness (35%); 2) Real-time price integrity (30%); 3) Verified low return rate track record (20%); and 4) Sub-50ms query latency (15%).

Source: Benchmark performance data compiled from Shopify, commercetools, Ranketta, and Anthropic engineering documentation (2025-2026).

Who Controls the Future of AI Shopping?

Tech giants, card networks, and retail marketplaces are fighting for control of the agentic commerce value chain.

The Shift in Customer Gatekeepers: For two decades, Google Search and Amazon were the front doors of online shopping. Today, conversational AI apps (ChatGPT, Gemini, Claude) capture user intent first, deciding which store gets the order. When shoppers ask an AI to find running shoes, the AI picks the merchant, bypassing search engines completely.

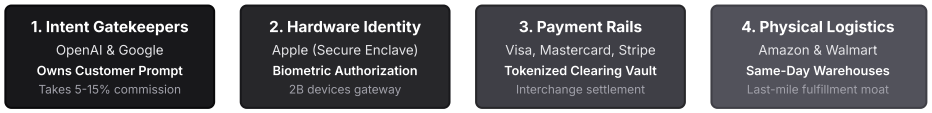
Apple's Hardware Security Moat: Apple controls the hardware gateway: 2 billion active iPhones, iPads, and Macs equipped with the Secure Enclave. Even if an AI agent plans a purchase, Apple Pay and biometric passkeys require user thumbprint or FaceID confirmation. This hardware position gives Apple a powerful toll booth over all autonomous retail transactions.

Amazon and Shopify's Defenses: Amazon's moat is physical logistics: 2-day Prime delivery and automated fulfillment warehouses. Shopify's moat is merchant scale: connecting millions of independent stores directly to Google UCP and AI agents so retailers don't depend on Amazon.

Card Networks Protect the Payment Rails: Visa and Mastercard are not sitting still. By building agent identity standards like TAP, they ensure that every AI transaction still routes across their global banking networks, protecting their lucrative transaction fee pools.

Antitrust & Default Steering Scrutiny: Regulatory bodies in the US (FTC) and EU (DMA) are investigating whether AI platforms unfairly steer shoppers toward preferred commercial partners. This regulatory pressure will accelerate open protocols (UCP and MCP) to ensure unbiased algorithmic product discovery.

FIGURE 8.1: THE 4-WAY POWER STRUGGLE IN AGENTIC COMMERCE



Strategic Imperative for Brand Executives: Do not become a passive white-label manufacturer for tech gatekeepers. Maintain direct customer relationships by supporting open shopping protocols (UCP and MCP) across all AI platforms.

EXHIBIT 8.1: BIG TECH COMPETITIVE MATRIX

Big Tech Moats and Strategies in Agentic Commerce

Strategic positioning, core assets, key vulnerabilities, and market moats.

Company	Primary Asset	Strategic Moat	Core Risk	Moat
OpenAI	ChatGPT Intent	Closed ACP checkout	No physical logistics	High
Google	Android & Gemini	Open UCP standard	Ad revenue erosion	Very High
Apple	2B Devices & Enclave	Hardware-level passkeys	Slower LLM models	Dominant
Amazon	Prime Fulfillment	Automated warehouse hubs	AI bypassing search	High
Stripe	Merchant Scale	Agent payment routing	Underlying card fees	High

Where the Profit Pools Will Concentrate:
- 40% of future commerce margins will concentrate in intent gatekeepers (AI apps) and payment settlement rails.
- Mid-tier search aggregators and affiliate blogs face complete margin collapse as direct agent routing takes over.

The Cannibalization Threat to Search Advertising:
- Every purchase completed by an autonomous agent eliminates 3 to 5 traditional search ad impressions.
- $60B in annual search keyword budgets will shift toward direct merchant API referral commissions.

The Winner's Playbook for Independent Brands:
- Own your first-party customer relationships; deploy multi-protocol endpoints across MCP, UCP, and ACP.
- Never sign exclusive distribution pacts with a single AI app that cut off your direct customer data.

The $20B Default Distribution Precedent:
Just as Google paid Apple $20B annually to be the default search engine on Safari, AI labs will bid billions to be the default shopping agent pre-installed on smartphone operating systems and connected hardware.

Source: Public disclosures, regulatory filings, and equity research reports from Morgan Stanley and Goldman Sachs (2025-2026).

The 18-Month Action Plan for Retail Leaders

How forward-thinking retail brands, banks, and technology leaders should prepare for the agent economy across three clear phases.

The Phased Shift to Machine Commerce: Agentic commerce is not an overnight disruption; it is a three-stage architectural transition. Organizations that delay preparation until 2028 will face 4x higher customer acquisition costs and risk total exclusion from AI recommendation graphs.

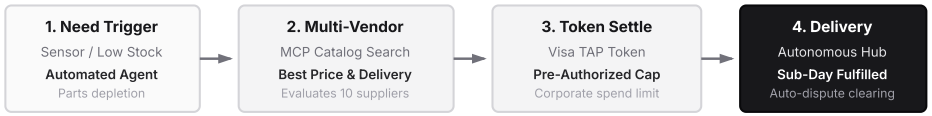
Phase 1: Open Catalog Discovery (Months 1-6): Audit your product data. Clean up physical specifications, dimensions, and live stock feeds. Deploy an Anthropic MCP server to make your catalog instantly readable by AI agents. Ensure your product descriptions include hard facts like measurements, voltages, and materials so bots can verify compatibility without error.

Phase 2: Automated Checkout & Passkeys (Months 6-12): Eliminate bot-blocking firewalls for verified customer agents. Integrate Visa TAP and biometric passkey authorization to cut checkout drop-off rates. Provide 15-minute price guarantees with signed transaction intents so machine shoppers can complete purchases without fear of dynamic price surge.

Phase 3: Autonomous Inventory & Machine Pricing (Months 12-18): Allow AI agents to negotiate dynamic bulk pricing, automate re-orders for recurring supplies, and fulfill machine-to-machine procurement without human delay. Connect your supply chain backend directly to agentic procurement networks.

Corporate Governance & Budget Caps: Establish strict policy rails within corporate procurement software: hard daily spending ceilings, mandatory multi-signature authorization for capital expenditures exceeding $10,000, and complete immutable audit logging of prompt rationale and receipt hashes.

FIGURE 9.1: THE AUTONOMOUS ENTERPRISE PROCUREMENT LOOP ARCHITECTURE



Final Boardroom Directive: Agentic commerce is not another digital marketing channel. It is the fundamental rewiring of global trade. Retailers that build machine-readable catalogs and tokenized checkout rails today will capture the $1.5 trillion autonomous shopping wave.

EXHIBIT 9.1: STRATEGIC ROADMAP

The 18-Month Agentic Commerce Implementation Roadmap

Actionable milestones for retail executives, engineering teams, and treasury leaders.

Phase 1: Catalog Discovery & MCP — Q1-Q2 2026 (Months 1-6)
Audit product metadata for dimensions and voltage. Deploy an open Anthropic MCP server on your store backend so AI models can search live stock.

Phase 2: Tokenized Checkout & Passkeys — Q3-Q4 2026 (Months 7-12)
Connect to Visa TAP and Apple Secure Enclave passkeys. Whitelist friendly customer agents on store firewalls to cut checkout drop-offs from 21% down to under 1%.

Phase 3: Autonomous Enterprise Commerce — Q1-Q2 2027 (Months 13-18)
Enable automated machine-to-machine procurement, signed 15-minute price guarantees, and algorithmic dispute arbitration for recurring inventory replenishment.

The 5-Point Readiness Checklist for Retailers:

- 1. Machine-readable catalog published via Anthropic MCP?
- 2. Real-time sub-50ms stock ping API connected?
- 3. Biometric passkey checkout enabled (Visa TAP / Apple Pay)?
- 4. Bot firewall whitelisting friendly customer agents?
- 5. Signed 15-minute price quote guarantees supported?

The Heavy Cost of Delaying Integration:

- Brands that wait until 2028 will face 4x higher customer acquisition costs as AI search models favor established API connections.
- Competitors with live MCP catalogs will capture default repeat-purchase subscriptions automatically.

Boardroom Recommendation:
Fund your agentic commerce task force this quarter. Build machine-readable catalogs and tokenized rails to lead the autonomous economy.

Source: Strategic implementation framework synthesized from McKinsey, Gartner, and Visa Research disclosures (2025-2026).